Kerr Landau-Siegel proof evaluation, Step 5: plausibility and context

Document evaluated: Bryce Kerr, "A simpler proof that there are no Landau–Siegel zeros," arXiv:2610.21099v1 [math.NT], 15 October 2026.

Built on: Steps 1 to 4 of this evaluation, files dated 18 Oct 2026, 1708, 1746, 1752 and 1758, all v1.

Also read in full for this step: - OpenAI, "Uniform exclusion of Landau–Siegel zeros," manuscript dated 1 October 2026, 9 pages (Kerr's reference [3]). - W. D. Banks, "On the OpenAI proof that Landau–Siegel zeros do not exist," arXiv:2610.15483v1, 11 October 2026, 19 pages (Kerr's reference [1]).

How to read this step. It is judgment, not checking, so every claim carries a band:

Band	Meaning
Band 1	Shown in Steps 1 to 4, or read directly in a named source.
Band 2	My reading or my own derivation, for an expert to check.
Band 3	Heuristic.

This step adds no breaks and removes none. The Step 4 tally stands: 1 false step, 32 breaks (7 substantive, 25 routine), and 3 notation defects.

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1. "Proves too much" tests

1.1 What the theorem would imply

Consequence	Status	Band
No Landau–Siegel zero for any real primitive χ: $(1-\beta)\log q \geq c$	Open since Landau, Page and Siegel (1930s). Banks, p. 2, calls it "one of the central problems of analytic number theory."	1 (Banks, p. 2)
An **effective** form, if c is computable (Section 2)	Siegel's $L(1,\chi) \gg_\varepsilon q^{-\varepsilon}$ is ineffective. Banks, p. 2: "the work of Goldfeld [2] and of Gross and Zagier [3] yields lower bounds of logarithmic order for the class numbers of imaginary quadratic fields, which is far from what the nonexistence of Landau–Siegel zeros would give."	1 (Banks, p. 2)
$L(1,\chi) \gg 1/\log q$ with a computable constant, so $h(-q) \gg \sqrt{q}/\log q$ effectively, by Dirichlet's class number formula	The standard deduction from a logarithmic zero-free interval at $s = 1$. I have not checked it against a source.	2
Effective versions of results currently stated "unless a Siegel zero exists" (for example, uniform estimates for primes in arithmetic progressions)	OpenAI, p. 1, and Banks, p. 2, both name these.	1 (as stated by those sources)
Conditional theorems that *assume* a Siegel zero become vacuous. The best-known example is Heath-Brown's result on twin primes under that assumption (I have not checked its date against the source).	Known in the field. A search-result snippet of a post by J. D. Lichtman makes the same point, but I could not read the post itself.	2

Test result (Band 2). Every consequence is believed true. All of them follow from the Generalized Riemann Hypothesis. None contradicts a known theorem or computation. So the theorem does not prove too much in the logical sense. It does claim a great deal: any one of the effective consequences alone would be a major result.

1.2 The algebraic core is unconditional, and it does not overreach

This is the sharpest test available for this paper, and it can be made quantitative.

Where the zero enters (Band 1). It enters only through Lemma 1 (Step 2, A5). Everything from Lemma 2 to the end is unconditional. Read without the zero, the argument proves this:

> **(Core.)** For $|q|$ large, with $N = |q|^\gamma$ and $Q \approx N^{4/3}/(\log N)^2$, the primes $p \in S$ (those with $\chi(p) = -1$ and $8 < p \leq Q$) satisfy $\sum_{p\in S} (\log p)/p \leq (13/10)\log B$, where $\log B \approx (1 + 1/(2\gamma))\log N$.

Why the core does not prove too much (Band 2, my analysis). - The total log-mass $\sum_{p\leq Q} (\log p)/p$ is about $\log Q \approx (4/3)\log N$. - So the core says the primes with $\chi(p) = -1$ carry at most a fraction of about $(13/10)/(4/3) =$ **39/40** of the log-mass up to Q, and in the limit the requirement tends to **15/16** (from 5/4 divided by 4/3). - Without an exceptional zero, the primes with $\chi(p) = -1$ are expected to carry about **1/2** of that mass (Band 3: the equidistribution heuristic). That is far below 39/40, so the core contradicts nothing known. - It also contradicts no actual q. To force $\chi(p) = -1$ for every prime up to X by

the Chinese remainder theorem, |q| must be exponentially large in X. Kerr's range, Q = |q|^{about (4/3)γ}, is vastly larger than log |q|.

A check on the method's own reach (Band 2, my analysis). - Kerr uses only primes whose Frobenius in K is σ or στ (χ(p) = −1). - Primes with trivial Frobenius (χ(p) = 1, (2/p) = 1) satisfy θ^p ≡ θ (mod pO). The analogue of Lemma 2 with x_1^p − x_1 is also unitriangular in L_D: by hand, because each extra monomial has a smaller x_1 exponent, and by machine for p ∈ {11, 13, 17} and D ∈ {50, 120}. So those primes could be used too. - Even so, the usable fraction without a zero would be about 3/4, and 3/4 × 4/3 = 1, still below 5/4. - The exponent 4/3 cannot be raised: Masser notes that his Theorem 1 is essentially best possible (p. 82), and μ ≤ rank Γ / n = 4/3 here (Lemma 7 together with the case L = {0}). - So the method **cannot** reach a contradiction unconditionally. It needs the exceptional zero to push the usable mass above 15/16.

This is the behavior a correct argument should show. An argument that produced a contradiction unconditionally would be wrong, since it would show no field K exists.

1.3 Does the argument depend on the target being a genuine L-function?

Yes, at two points. - **Analytic (Band 1):** Lemma 1 rests on Davenport's (2), which needs the Euler product and the functional equation of a primitive L(s, χ) (Step 4, Section 2). - **Algebraic (Band 1):** The p-adic step needs a genuine Frobenius action, (√q)^p ≡ χ(p)√q (mod pO). That ties the coefficients χ(p) to the splitting of primes in Q(√q).

A Davenport–Heilbronn-type function has zeros off the line, but it has no Euler product, so neither step can even be stated for it (Band 2). This test does not fire.

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2. Effectivity

Ingredient	Kerr	Banks (OpenAI argument)	Band
Analytic constant in Lemma 1	Davenport's absolute constants c_1 and c_2, never evaluated. They are effective in principle.	Explicit: $C_1 = 2e^{3/2}/9 \approx 0.9959$ and $C_2 = e^{3/2}/4 \approx 1.1204$ (Lemma 2.1, Remark 7.1), using Rosser and Schoenfeld's explicit Mertens bound.	1 (Banks), 2 (that Davenport's constants are effective)
Mertens and Chebyshev	Used silently (Step 2, B27, A30)	Rosser–Schoenfeld; Σ_{p≤M} log p ≤ 2M	1
Threshold for "N large"	Implicit. Step 3 found it small: (11) held at every grid point from N = 200 to N = 10^7.	Explicit	1 (Step 3)
Large-q constant	Not given	$q_0 = 10^{10}$, $\delta_0 = 3.0 \times 10^{-3}$	1 (Banks, Theorem 1.1, Remark 7.1)
Small q	Missing (Step 2, B2)	A computation by Lu, Zaman and Zhao (arXiv:2602.03626), which Banks calls "a computation that we have not verified"	1 (Banks, Remark 7.1)

Verdict (Band 2). Kerr's argument is effective in principle: every constant it uses can be computed, and Part C of Step 2 makes the parameter choices explicit in terms of C_1. Kerr does not compute c, or say that it can be computed.

Banks's version of the OpenAI argument does give explicit constants. Combined with the small-conductor computation, which neither Banks nor I have checked, that would give $c = 3.0 \times 10^{-3}$ for every q ≥ 3.

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3. How the method gets past the known obstacles

3.1 The mechanism

(Band 1, from Steps 1 to 4.) 1. Adjoining √2 to Q(√q) gives a rank-4 lattice of points (θ_n, σθ_n, στθ_n) in C^3. 2. At a prime with χ(p) = −1, Frobenius sends θ to σθ or στθ, so x_1^p ≡ x_2 or x_3 at those points. A polynomial built from the factors (x_1^p − x_ℓ) therefore takes values divisible by high powers of p. 3. Pigeonhole (Siegel's lemma) gives a polynomial of degree about N^{4/3} that vanishes modulo P at all N^4 points. A size bound and the norm then force exact vanishing. 4. Masser's zero estimate says nothing of that degree can vanish there.

(Band 2.) This is the auxiliary-polynomial method of transcendence theory, not an analytic one. The exceptional zero is used only to guarantee enough primes with χ(p) = −1.

3.2 Why the classical obstacles do not apply

(Band 2.) - **What the classical methods see.** They work through the zeros of L-functions: zero-free regions, zero-density estimates, and the Deuring–Heilbronn phenomenon. A single real zero is consistent with every averaged statement. Siegel's

theorem is ineffective because it can only compare two hypothetical exceptional characters. - **What this argument uses instead.** It never looks at other zeros. It turns a Siegel zero into the equivalent statement that $\chi(p) = -1$ for almost all primes up to q^A (compare OpenAI's citation of Friedlander and Iwaniec, Section 5). It then refutes that statement with algebraic arithmetic: integrality, the norm, Frobenius congruences, and a zero estimate. - **Precedent.** This is the same kind of escape as the Goldfeld and Gross–Zagier route: arithmetic input from outside analytic number theory. - **Uniformity, the property such arguments most often lose.** Kerr's version gets it from the **statement** of Masser's theorem: (M/n)^μ involves no constant depending on the points (Step 4, Section 1.1). OpenAI's version gets it from a self-contained interpolation lemma whose constant, 32, is "independent of A and H" (OpenAI, Corollary 4). Step 4 confirmed that Kerr's application meets Masser's hypotheses exactly.

3.3 Kerr against OpenAI and Banks

Element	OpenAI (1 Oct) and Banks (11 Oct)	Kerr (15 Oct)	Band
Analytic input	Lemma 2 / Lemma 2.1, the same inequality as Kerr's Lemma 1, with the derivation on the page	Lemma 1, derivation omitted (B4)	1
Field	Q(√d, √2) with d squarefree. d = 2 is excluded explicitly: "except for one fixed quadratic field" (OpenAI, p. 1), and "the case d = 2 occurs only for q = 8, so it is excluded when q ≥ 9" (Banks, §1.5)	Q(√q, √2) with q the discriminant, and "degree four" stated without exception	1. **Kerr's false step FS1 is absent from both predecessors.**
Frobenius congruence	Proved by Euler's criterion (OpenAI §5; Banks Lemma 6.1)	Cited to Milne (B7)	1
Auxiliary object	An interpolation determinant: a weighted greedy selection of rows, Hadamard's inequality, and divisibility of the determinant by p^{E_p}	A pigeonhole polynomial F ∈ L_D vanishing modulo P, then exact vanishing via the norm	1
Zero estimate or interpolation	Proved on the page (OpenAI Lemma 3; Banks Proposition 3.4, about 4 pages of convex geometry and linear algebra)	Imported: Masser 1981, 15 pages of commutative algebra (Nesterenko-type ideal theory, unmixedness, degrees)	1
Exponent 4/3	U = N^{4/3} with $U^3 = N^4$	μ(Γ) ≥ 4/3 (Lemma 7)	1
Margin	Banks: 13/16 against 1, with explicit constants	5/4 < 13/10 < 4/3	1
Small q	OpenAI: by contradiction along a sequence. Banks: explicit, plus the Lu–Zaman–Zhao computation	Missing (B2)	1
AI involvement as stated	OpenAI: an unreleased internal model (secondary source, Section 4). Banks: Claude produced the initial report and expanded draft; Banks verified every line.	"ChatGPT assisted with the mathematical development." Banks shared a draft with Kerr.	1

"Simpler" (Band 2). Kerr's paper is shorter on the page because it imports Masser. It is not simpler in its foundation. Masser's proof uses heavier machinery (primary decomposition, unmixedness, Cohen's theorem on regular local rings, and Gröbner's theory of degree) than OpenAI's elementary interpolation lemma, which Banks writes out in full. Kerr's real simplification is replacing the determinant and row selection with pigeonhole. Measured by our standard, Kerr's text has more breaks than Banks's: for example, B4, B2 and FS1, which Banks's text does not have.

3.4 Relation to our Quasi-Riemann Hypothesis evaluation

(Band 1, from that evaluation's Step 6 report, 7 Oct 2026, v2.) - That evaluation covered a different OpenAI paper: 199 pages, claiming no zeros with Re s > 7/8 for all Dirichlet and Hecke L-functions over Q(√−3). - Its result would also exclude Siegel zeros. - That proof rests on Patterson's cubic theta function, and its risk was concentrated in Lemmas 17.1 and 18.1, which we could not rebuild independently.

(Band 2.) The Siegel-zero argument here is entirely separate. It uses no theta function and no zero-free half-plane, and its correctness does not depend on the 7/8 paper. It proves much less (only real zeros near 1). It is also short enough to check completely, which is what Steps 1 to 4 have done. The 7/8 result would imply this one; the converse does not hold.

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4. Outside reaction

Caution. Except where noted, what follows comes from web-search summaries and from secondary pages read through a fetch tool. None of it is a primary statement I have read in full. None of it bears on whether the mathematics is correct.

Finding	Source	Band
Banks is the one named mathematician I found publicly stating that he believes the argument is sound. He "worked through the argument line by line" and writes: "In view of the importance of the statement, independent scrutiny is essential."	Banks, p. 2, read in full	1
A problem-tracking site lists the Landau–Siegel entry as "Candidate (review pending)," with "Lean-checked, statement unaudited" and "No independent mathematician has checked this yet" (entry added 7 October 2026, before Banks and Kerr).	vibemathed.com, via fetch summary	3
A secondary article quotes Alex Kontorovich and Hector Pasten with very strong reactions (Pasten: the result "solved the problem"). The context and original wording are not confirmed.	36kr.com, via fetch summary	3
A post by J. D. Lichtman refers to "the Siegel zero dispr[oof]" and to conditional work that assumed a Siegel zero.	Search-result snippet only; the post itself was not read	3
The broad reaction to OpenAI's 6 October release concerns how it was released, not this proof. Examples are the AHM statement reposted on Terence Tao's blog (7 October) and Fortune (7 October). Neither mentions the Landau–Siegel manuscript. Fortune reports that OpenAI withdrew solutions to at least three other problems.	Tao blog and Fortune, via fetch summary	3
I found **no published refutation** of the OpenAI, Banks or Kerr arguments, and no named expert reporting an error. This is an absence in what the search found, not evidence of correctness.	Searches on 18 October 2026	3

Pages I could not read: - The arXiv abstract page for Kerr's paper (arxiv.org/abs/2610.21099). The fetch was rate-limited, so I could not check for a revised version or a comment field after v1. - The *Science* article "AI edges closer to solving the Riemann hypothesis." The fetch returned an error.

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5. Overall assessment

1. **On the page (Band 1).** Kerr's paper is not a complete proof by our standard. It has 1 false step (q = 8), 32 breaks (7 substantive) and 3 notation defects, and its foundation is unstated.
2. **Above the foundation (Band 1).**
   - I found no mathematical defect beyond FS1, which is trivially repaired.
   - Every break can be filled. Step 2 Part C and Step 4 fill them all.
   - Every number checks by machine (Step 3).
   - Every citation supports its use once the bridging steps are supplied (Step 4).
3. **Plausibility (Band 2).**
   - The argument passes both "proves too much" tests.
   - It uses the exceptional zero in exactly the way a correct argument must: it cannot reach a contradiction without the zero, because the usable prime mass would be about 1/2 to 3/4 against a requirement of 15/16.
   - Its uniformity in q comes from a published theorem whose hypotheses its application meets exactly.
4. **Independence (Band 3).**
   - Three write-ups agree: OpenAI, Banks and Kerr. But they share the central idea (the √2 adjunction, the Frobenius substitution and the 4/3 exponent), and Banks shared his draft with Kerr. So they are not three independent confirmations.
   - Two used AI assistance: Banks used Claude, and Kerr used ChatGPT.
   - This evaluation was also done by Claude. Errors common to these systems would not be caught by any of them. The evaluation is therefore not a substitute for review by expert number theorists and transcendence theorists.
5. **Base rate (Band 3).** Claimed solutions to famous problems usually fail. What sets this case apart:
   - the argument is short;
   - the import that carries it (Masser) is a published 1981 theorem;
   - its arithmetic margin is a clean constant (1/30 in Kerr; 3/16 in Banks), not a thin numerical one.

   The plausible places for a hidden error are the ones Steps 2 and 4 have already examined most closely:
   - the uniformity of the zero estimate in the lattice;
   - the passage from vanishing modulo P to exact vanishing;
   - the order in which the parameters are chosen.

   None failed.

Questions to put to the author, in priority order

1. State the foundation up front, and state Lemma 6 for a basis, or cite Masser's Proposition for the generator form (B1, B22).
2. Correct "The field K has degree four" for q = 8, or restrict to |q| > 8 (FS1).
3. Derive Lemma 1 on the page, including the unnumbered ζ'/ζ bound and the conductor fact (B4, F13, F15).
4. Fix the order of quantifiers (γ, then ε, then q_0) and execute the inequality with the 1/30 margin (B26, B28).
5. Derive the asymptotics for r, W and E_p and execute "(9) implies (11)" (B20, B21).
6. Close the small-|q| case and name c. State whether c is effective (B2, Section 2).

Proposed Step 6: final shareable report

A single self-contained document combining Steps 1 to 5. It would contain: - a one-page summary for a non-specialist reader; - the verdict by band; - the false step; - the 32 breaks in one table, with severity and the repair for each; - the citation findings; - the machine-check results, with the scripts referenced by file name; - the comparison with OpenAI and Banks; - the questions for the author; - the full list of sources.

Mathematics in the derivations would stay labeled as mine, never credited to Kerr. The output would be "Step 6 - Kerr Landau-Siegel proof evaluation Final Report 18 Oct 2026 HHMM_v1.md", saved in the folder.

Sources

Primary, read in full: - Kerr, A simpler proof that there are no Landau–Siegel zeros (arXiv:2610.21099) - Banks, On the OpenAI proof that Landau–Siegel zeros do not exist (arXiv:2610.15483) - OpenAI, "Uniform exclusion of Landau–Siegel zeros," manuscript dated 1 October 2026 (Arnoldo's copy) - Masser, Invent. Math. 63 (1981); Davenport, *Multiplicative Number Theory*, 2nd ed. (1980); Milne, *Algebraic Number Theory* v3.03 (Arnoldo's copies)

Secondary, read through fetch summaries (Band 3): - VibeMathed: The Landau–Siegel zero problem - 36kr: "Bombshell Breakthrough! After OpenAI, AI Has Once Again Cracked the Generalized Riemann Hypothesis" - AHM Statement on OpenAI's October 6 Release (Terence Tao's blog) - Fortune, 7 October 2026 - AI Signal: OpenAI publishes 722 AI-written math manuscripts - J. D. Lichtman on X (search snippet only) - Woit, "No Landau-Siegel zeros?" (2022). This turned out to be about Yitang Zhang's 2022 claim, not this one. It is listed only because the search returned it.

Could not read: arXiv abstract page for 2610.21099 (rate-limited); Science article (fetch error).